

UNIT 4 – Web Application & Usability Testing

Exhaustive Study Notes covering UI/UX, Browser Matrix, Form Validation, Links & Responsive Design

1. Web Application Testing Overview

Web Application Testing tests websites hosted on web servers across client browsers. It verifies UI layout, application business rules, security, performance, and multi-device usability.

3-Tier Web Application Testing Architecture



Diagram: 3-Tier Web Application Testing Architecture.

2. UI & UX Testing

- User Interface (UI) Testing: Verifies graphic elements (fonts, colors, alignment, buttons, images, margins).
- User Experience (UX) Testing: Evaluates how user-friendly, intuitive, and efficient the user workflow is.

3. Browser Compatibility & Cross-Browser Testing

Cross-Browser Testing ensures web apps render identically across different browser engines:

- Blink Engine: Google Chrome, Microsoft Edge, Brave, Opera.
- Gecko Engine: Mozilla Firefox.
- WebKit Engine: Apple Safari.

Testing Matrix: Verify layouts on Chrome, Firefox, Edge, Safari across Windows, macOS, iOS, Android.

4. Form Testing & Input Validation

Web forms are primary entry points for user data. Testing includes:

- Mandatory Field Checks: Submit form with empty required fields.
- Data Boundary Limits: Enter strings shorter/longer than max length.
- Special Character Sanitization: Prevent XSS scripts (`<script>alert(1)</script>`) and SQL Injection (`' OR '1'='1`).
- File Upload Boundaries: Validate file formats (.pdf, .jpg), file size limits (5MB), and corrupt file handling.

Web Form Input Validation & Security Lifecycle

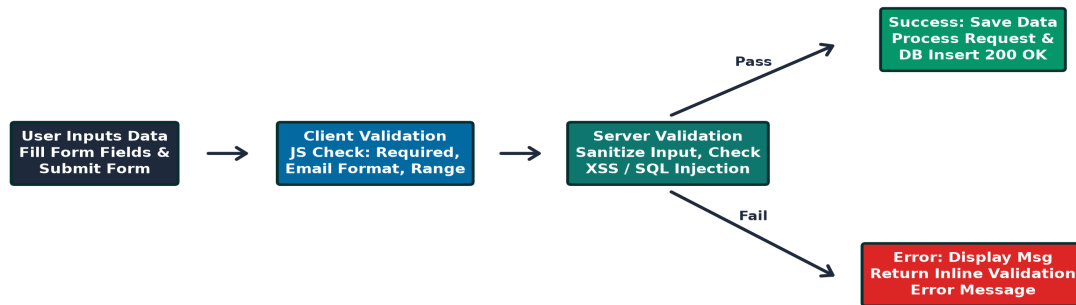


Diagram: Web Form Input Validation & Security Lifecycle.

5. Link Testing & Broken Links

Validates all links on web pages:

- Internal Links: Navigation between pages within website.
- External Links: Outbound links to external websites.
- Anchor Links: Jump links within the same page.
- Mailto / Tel Links: Triggers email client or phone call.
- Broken Link Check: Verify no link returns HTTP 404 Not Found error.

6. Responsive Testing & Viewport Breakpoints

Responsive Testing verifies application UI layout auto-adapts across screen dimensions:

- Desktop Breakpoint: $\geq 1200\text{px}$
- Tablet Breakpoint: $768\text{px} - 1024\text{px}$
- Mobile Breakpoint: $\leq 480\text{px}$

Key checks: Touch targets (buttons $\geq 44 \times 44\text{px}$), text readability without horizontal scroll, mobile menu drawer toggle.

Cross-Browser & Responsive Breakpoint Matrix

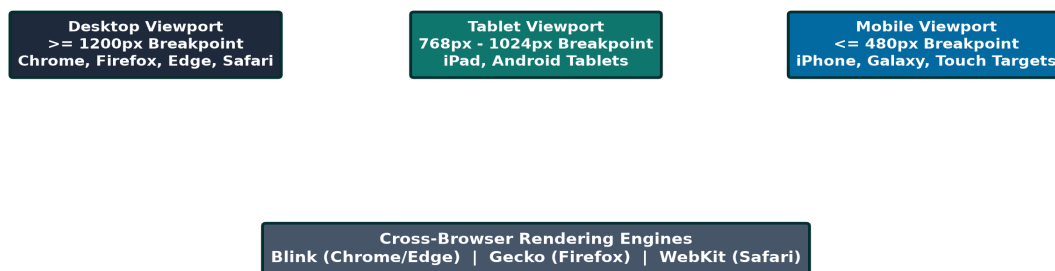


Diagram: Cross-Browser & Responsive Breakpoint Matrix.